

# \* Exceptional Decider : Probability Analysis Report

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## 1. Understanding Basics

→ what is probability?

Ans → Probability is a number between 0 and 1 that measures how likely an event is to happen.  
0 means it cannot happen, 1 means it is certain.

→ Key terminology

- Ans →
- Experiment :- a process with an uncertain result (picking a student at random).
  - Outcome :- One possible result (the student passed).
  - Sample Space (S) :- all possible outcomes. Here, the 200 students.
  - Event :- a subset of the sample space (students who attend more than 80%).
  - Complement :- ( $A'$ ) everything not in A, so  $P(A') = 1 - P(A)$ .
  - Joint probability  $P(A \cap B)$  :- Both event happens. Union  $P(A \cup B)$  :- at least one happens.
  - Marginal probability :- the probability of one event on its own.
  - Conditional probability  $P(A|B)$  :- probability of A given B has occurred =  $P(A \cap B) / P(B)$ .

→ These probability events the dataset.

| Event                                        | Count | probability       |
|----------------------------------------------|-------|-------------------|
| E1: Student passes the exam                  | 109   | $109/200 = 0.545$ |
| E2: Student participates in group discussion | 110   | $110/200 = 0.550$ |
| E3: Student studies more than 10 h/week      | 130   | $130/200 = 0.650$ |
| E4: (Bonus) Attendance above 80%             | 83    | $83/200 = 0.415$  |

## 2. Types of Events

- Empirical probability (based on observed data).

Ans → Scenario :- a randomly chosen student passes the final exam.

$$P(\text{Pass}) = \text{Students who passed} / \text{total students} = 109/200 = 0.545 \text{ (54.5\%)}$$

- Theoretical probability (based on equally likely outcomes).

Ans → Scenario :- the exam result has two outcomes, pass or fail.

$$P(\text{Pass}) = \text{favourable outcomes} / \text{total outcomes} = 1/2 = 0.500 \text{ (50\%)}$$

### 3. Random Variable and probability Distribution.

Ans → Let  $x$  = numbers of Students who passed the final exam out of 3 randomly selected students.  $x$  can take the value 0, 1, 2, 3.

Assumption independent selections (a reasonable approximation for 3 student from 200) with  $p = P(\text{pass}) = 0.545$  and  $q = 1 - p = 0.455$ ,  $x$  follows a binomial distribution  $B(n=3, p=0.545)$ .

$$P(x=k) = C(3, k) \cdot 0.545^k \cdot 0.455^{3-k}$$

| $x$   | Calculation                     | $P(X=x)$ | $x \cdot P(x)$ | $x^2 \cdot P(x)$ |
|-------|---------------------------------|----------|----------------|------------------|
| 0     | $1 \times 0.455^3$              | 0.0942   | 0.0000         | 0.0000           |
| 1     | $3 \times 0.545 \times 0.455^2$ | 0.3385   | 0.3385         | 0.3385           |
| 2     | $3 \times 0.545^2 \times 0.455$ | 0.4054   | 0.8109         | 1.6218           |
| 3     | $1 \times 0.545^3$              | 0.1619   | 0.4856         | 1.4568           |
| Total |                                 | 1.0000   | 1.6350         | 3.4171           |

→ Mean and Variance

Ans → Mean  $E(x) = \sum x \cdot P(x) = 1.635$

(check :-  $np = 3 \times 0.545 = 1.635$ ).

Variance  $\text{Var}(X) = E(x^2) - [E(x)]^2 =$

$3.4171 - 1.635^2 = 3.4171 - 2.6732 = 0.7439$

(check :-  $npq = 3 \times 0.545 \times 0.455 = 0.7439$ ).

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$$\text{Standard deviation} = \sqrt{0.7439} \approx 0.863$$

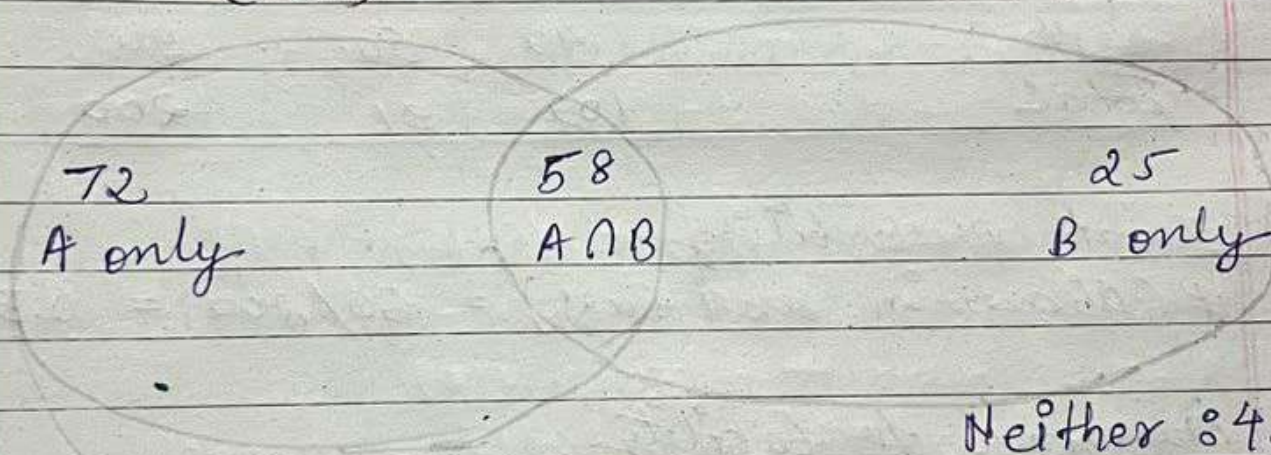
→ Interpretation :- Out of 3 random students we expect about 1.6 to pass. The most likely single outcome is exactly 2 passes (40.5%).

#### 4. Venn Diagram.

Ans → Let A = Students who study more than 10 hr/week and B = Students with attendance above 80%.

A : Study > 10 hrs  
(130)

B : Attendance > 80%  
(85)



| Region                               | Count | probability |
|--------------------------------------|-------|-------------|
| A only (Study > 10, attendance ≤ 80) | 72    | 0.360       |
| A ∩ B (Both Condition)               | 58    | 0.290       |
| B only (attendance > 80, study ≤ 10) | 25    | 0.125       |
| Neither                              | 45    | 0.225       |
| Total                                | 200   | 1.000       |

→ check with the addition rule :-

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.650 + 0.415 - 0.290 = 0.775, \text{ which equals } (72 + 58 + 25) / 200 = 155 / 200.$$

$$\text{Also, } P(B|A) = 58/130 = 0.446,$$

versus  $P(B) = 0.415$ , so students who study more attend slightly more often.

## 5. Contingency table and probability calculations.

Ans →

| Group discussion | Pass | fail | Row total |
|------------------|------|------|-----------|
| yes              | 69   | 41   | 110       |
| No               | 40   | 50   | 90        |
| Total            | 109  | 91   | 200       |

a) Joint probability

$$\rightarrow P(\text{Discussion and Pass}) = 69/200 = 0.345$$

b) Marginal probability

$$\rightarrow P(\text{Pass}) = 109/200 = 0.545$$

c) Conditional probability

$$\rightarrow P(\text{Pass} | \text{Discussion}) = P(\text{Discussion} \cap \text{Pass}) / P(\text{Discussion}) = 0.345 / 0.550 = 69/110 \approx 0.627.$$

6. Understanding Relationships  
Conditional probability in plain language.  
Ans → If I already know something about a student, how does the chance of the outcomes change?

Knowing a fact shrinks the group we look at. Instead of all 200 students,  $P(\text{Pass} | \text{Discussion})$  only looks at the 110 who participated and asks what share of them passed (69 of 110 = 62.7%). The formula

$P(A|B) = P(A \cap B) / P(B)$  simply says: of everyone in B, what fraction is also in A? Here the answer (62.7%) is higher than the overall 54.5% so knowing a student particular makes passing look more likely.

- Are "participating in group discussion" (D) and "passing the exam" (P) independent, dependent or mutually exclusive?

| Test                                                  | Calculation                                    | Result                              |
|-------------------------------------------------------|------------------------------------------------|-------------------------------------|
| Mutually exclusive<br>(needs $P(D \cap P) = 0$ )      | $P(D \cap P) = 69 / 200 = 0.345$               | Not zero, so not mutually exclusive |
| Independent? (needs $P(D \cap P) = P(D) \cdot P(P)$ ) | $0.550 \times 0.545 = 0.300 \text{ vs } 0.345$ | Not equal, so not independent       |
| Independent? (needs $P(P D) = P(P)$ )                 | $0.627 \text{ vs } 0.545$                      | Not equal, so not independent       |

7. Bayes' Theorem Application  
Ans → Let  $H$  = Students had high attendance  
( $> 80\%$ ),  $P$  = student passed,  $F$  =  
student failed, Given :-

$$P(H) \quad P(H|P) = 0.70$$

$$P(H|F) = 0.40$$

$$P(H) = 0.60$$

→ Goal :-  $P(P|H)$ .

→ Step :- 1 :-

find the prior  $P(P)$ . The pass rate isn't  
given directly, but it follows from  
the law of total probability. with

$$P(F) = 1 - P(P) :-$$

$$\bullet P(H) = P(H|P) \cdot P(P) + P(H|F) \cdot P(F)$$

$$\bullet 0.60 = 0.70 \cdot P(P) + 0.40 \cdot (1 - P(P))$$

$$\bullet 0.60 = 0.40 + 0.30 \cdot P(P)$$

$$\bullet P(P) = 0.20 / 0.30 = 2/3 \approx 0.667$$

$$\text{So } P(F) = 1/3.$$

→ Step :- 2 :-

$$\bullet P(P|H) = P(H|P) \cdot P(P) / P(H)$$

$$\bullet (0.70 \times 0.6667) / 0.60$$

$$\bullet 0.4667 / 0.60$$

$$\bullet 7/9 \approx 0.778 \quad (77.8\%)$$

→ check using the other branch :-

$$P(F|H) = (0.40 \times 1/3) / 0.60 = 0.222;$$

and

$$0.7778 + 0.2222 = 1$$

| Group | Prior  | P(H) group | Joint<br>P(H ∩ group) | Posterior<br>P(group   H) |
|-------|--------|------------|-----------------------|---------------------------|
| Pass  | 0.6667 | 0.70       | 0.4667                | 0.7778                    |
| Fail  | 0.3333 | 0.40       | 0.1333                | 0.2222                    |
| Total | 1.0000 |            | 0.6000                | 1.0000                    |

\* Final Summary :-

→ Using probability tools on the 200 students dataset, about 54.5% of students pass the exam. Passing is not ~~freedom~~ random chance: every factor we examined shifts the odds.

→ Previous test score is the strongest factor. Pass rate rises from 29% (Score below 50) to 85% (Score 70 or more).

→ Study hours come next. Student studying 16+ hrs pass 77% of the time against 30% for those under 8 hours.

→ Attendance matters most at the top end. The pass rate jumps to 86% above 90% attendance.